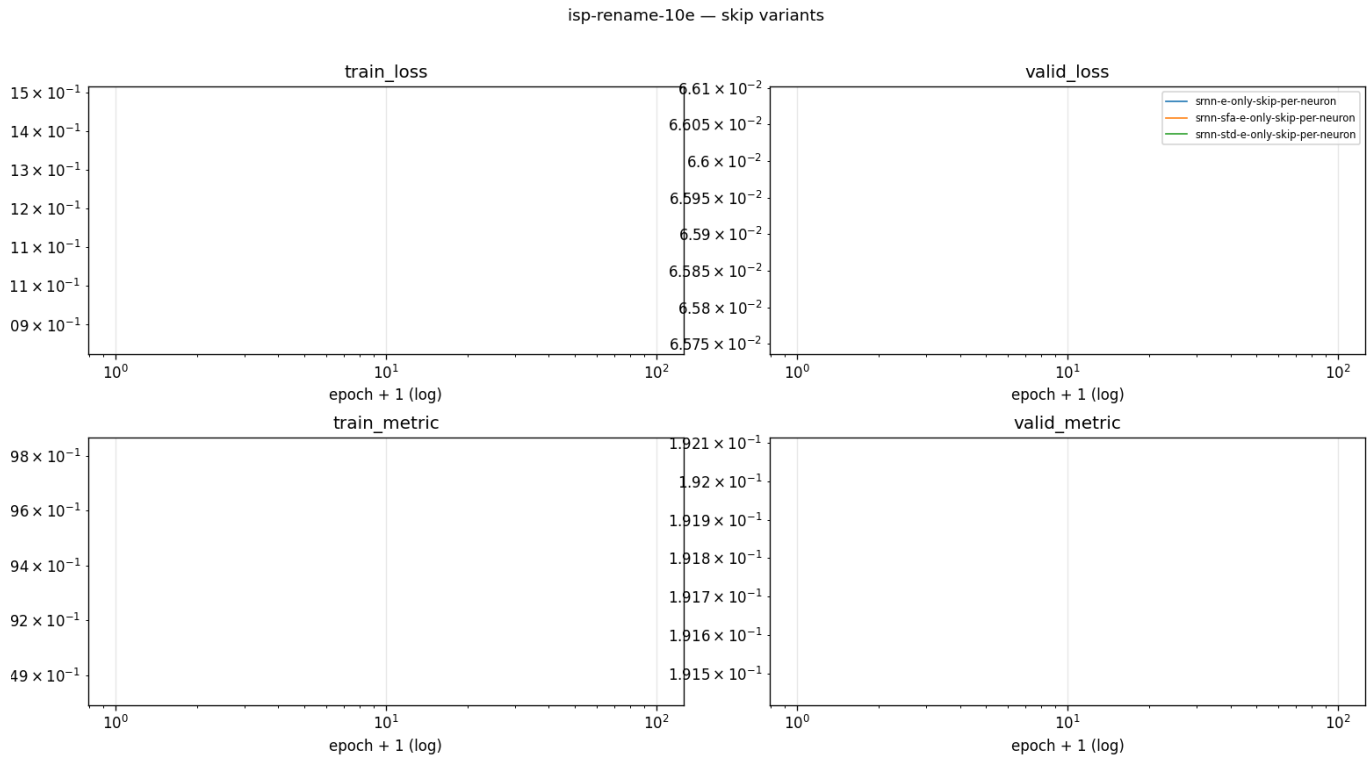


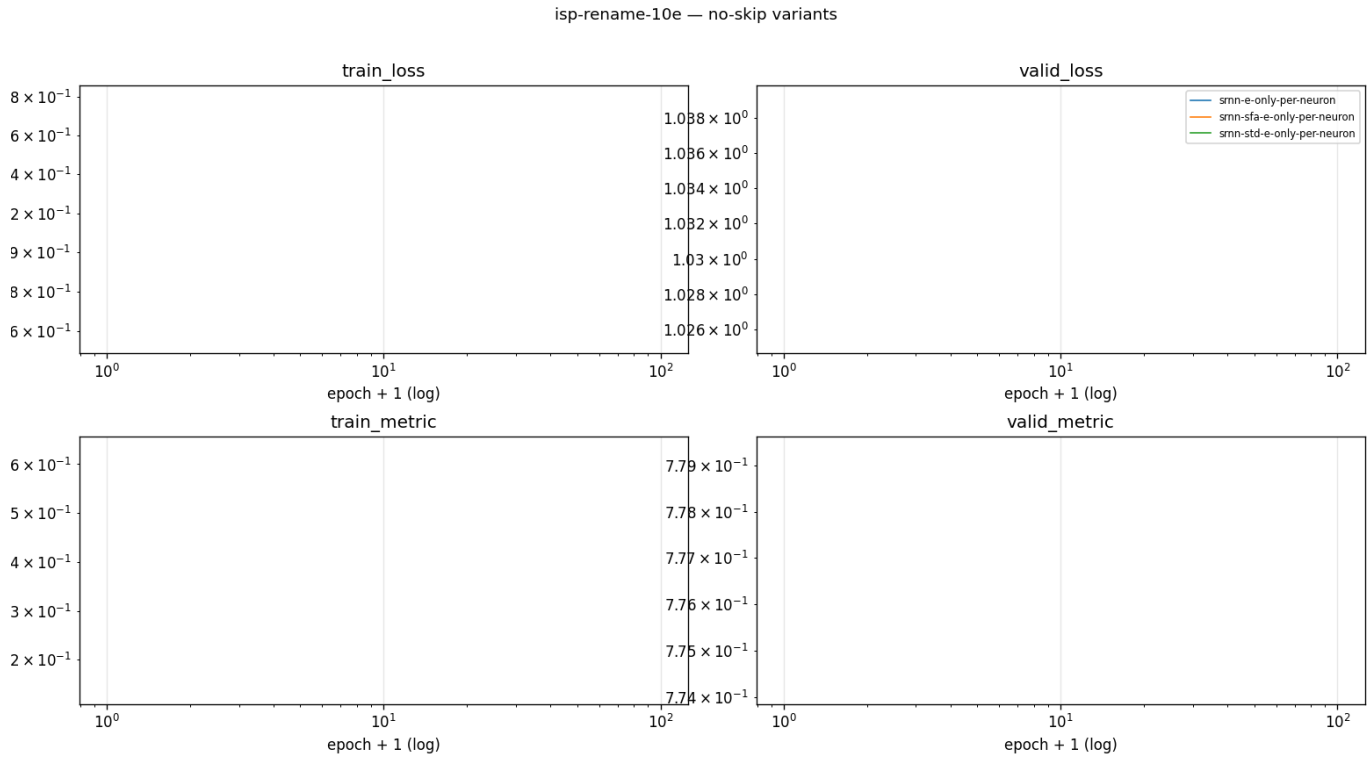
Run: isp-rename-10e

Log-log loss curves



Final test_loss (init - last) [skip]:
 srnn-e-only-skip-per-neuron: 0.06561 → 0.06492
 srnn-sfa-e-only-skip-per-neuron: 0.08121 → 0.06521
 srnn-std-e-only-skip-per-neuron: 0.06591 → 0.06493

Figure 1: skip variants — log-log loss/metric



Final test loss (init - last) [no-skip]:
 smn-e-only-per-neuron: 0.98043 - 0.89105
 smn-sfa-e-only-per-neuron: 0.91505 - 0.89835
 smn-std-e-only-per-neuron: 0.90077 - 0.89075

Figure 2: no-skip variants — log-log loss/metric

srnn-e-only-per-neuron

isp-rename-10e — effective time constants (srnn-e-only-per-neuron)

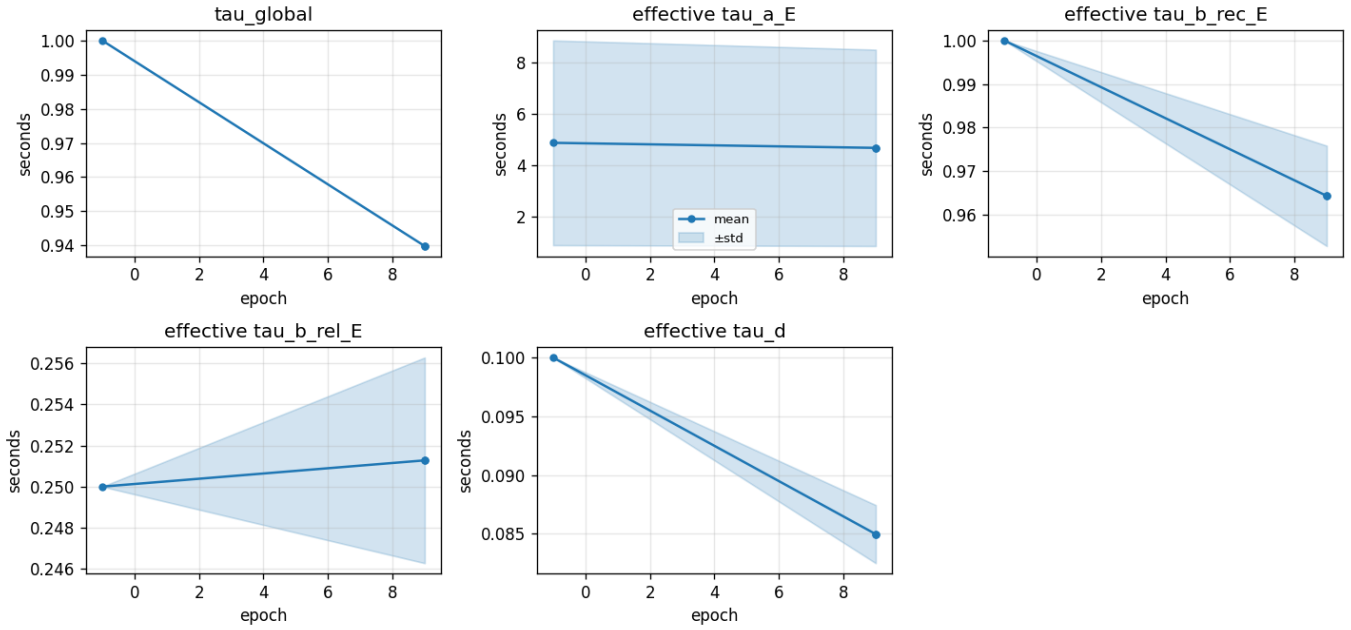


Figure 3: Tau evolution

isp-rename-10e — recurrent E vs I weights (srnn-e-only-per-neuron)

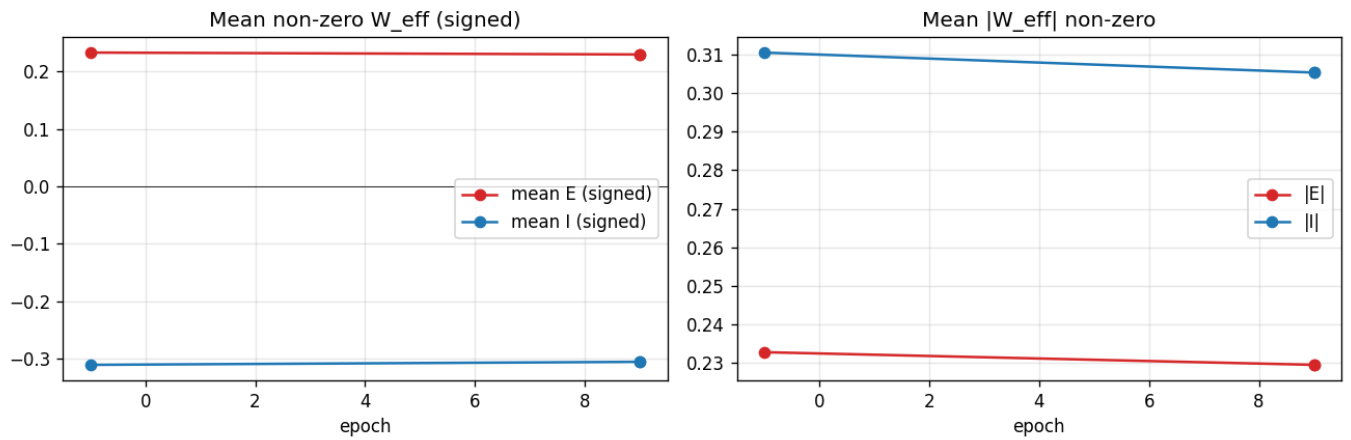


Figure 4: W_EI evolution

srnn-e-only-per-neuron — parameter table

isp-rename-10e - variant srnn-e-only-per-neuron (k=0)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.93981	-	-	scalar
tau_d (s)	+0.1	+0.084968	0	0.002474	300
tau_a_E[j=0] (s)	+0.25	+0.24145	0	0.003984	150
tau_a_E[j=1] (s)	+4.3832	+4.2053	0	0.006571	150
tau_a_E[j=2] (s)	+10	+9.5948	0	0.002447	150
tau_b_rec_E (s)	+1	+0.9643	0	0.01153	150
tau_b_rel_E (s)	+0.25	+0.25128	0	0.004998	150
W_eff (E src, signed)	+0.23267	+0.22938	0.07641	0.07542	15048
W_eff (I src, signed)	-0.31062	-0.30543	0.07724	0.07612	14977
W_eff (E src)	+0.23267	+0.22938	0.07641	0.07542	15048
W_eff (I src)	+0.31062	+0.30543	0.07724	0.07612	14977
W_in_eff (input neurons)	-0.0003386	-0.00033055	0.1012	0.1105	6675
c_E[j=0] (SFA coupling)	+0.05	+0.04879	0	0.0008542	150
c_E[j=1] (SFA coupling)	+0.05	+0.048835	0	0.000923	150
c_E[j=2] (SFA coupling)	+0.05	+0.048769	0	0.0009456	150
c_0_E[j=0] (SFA offset)	+0	-0.042226	0	0.02024	150
c_0_E[j=1] (SFA offset)	+0	-0.042622	0	0.01952	150
c_0_E[j=2] (SFA offset)	+0	-0.04238	0	0.01908	150
a_0 (threshold)	+0.35	+0.31537	0	0.024	300
readout_weight	-2.8854e-05	+0.00051796	0.06604	0.0711	6675
readout_bias	+0	-0.0015425	0	0.02677	89
ic.ic (per-variant init)	+0.10386	+0.10386	0.3828	0.3828	900

srnn-e-only-skip-per-neuron

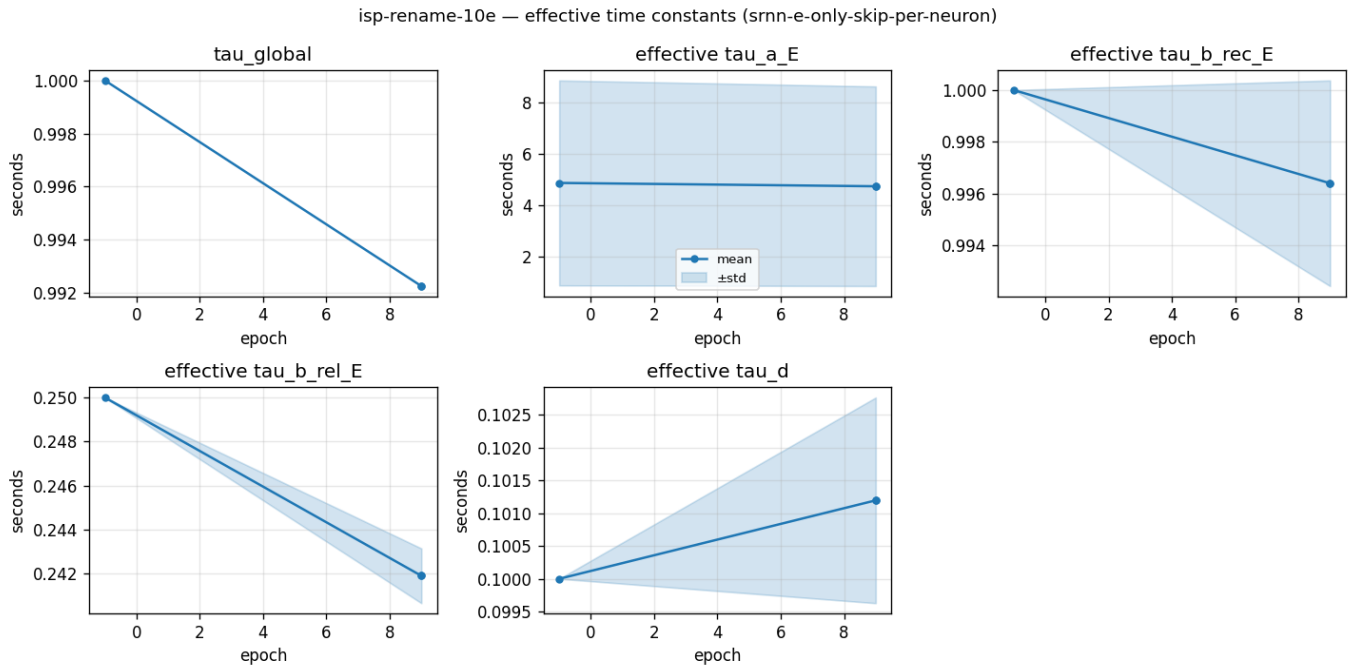


Figure 5: Tau evolution

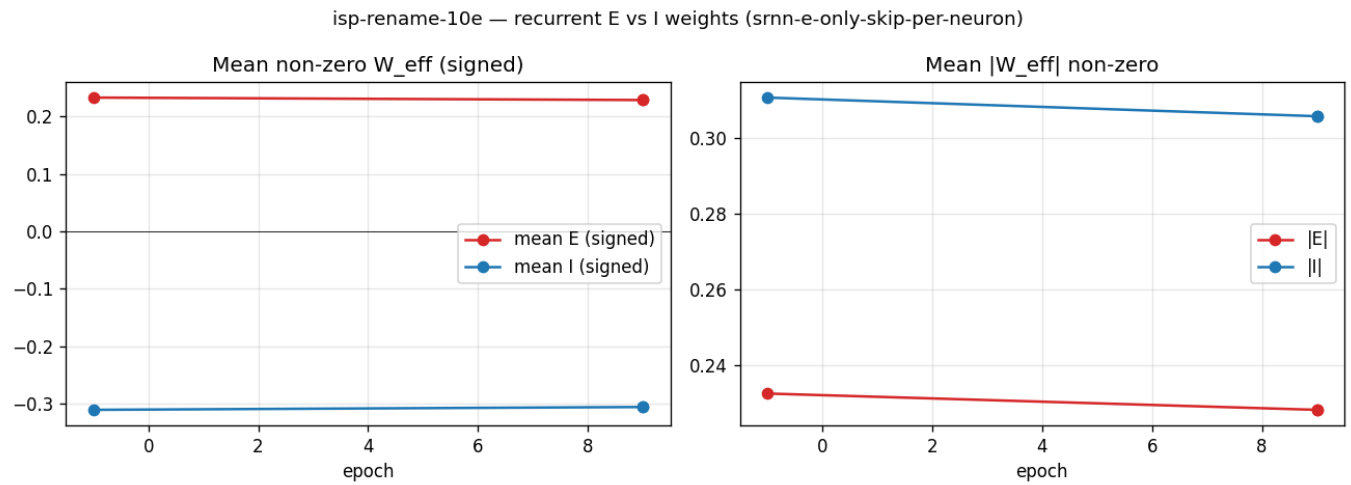


Figure 6: W_EI evolution

srnn-e-only-skip-per-neuron — parameter table

isp-rename-10e - variant srnn-e-only-skip-per-neuron (k=1)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.99226	-	-	scalar
tau_d (s)	+0.1	+0.10119	0	0.001566	300
tau_a_E[j=0] (s)	+0.25	+0.24083	0	0.001541	150
tau_a_E[j=1] (s)	+4.3832	+4.265	0	0.002706	150
tau_a_E[j=2] (s)	+10	+9.7362	0	0.001039	150
tau_b_rec_E (s)	+1	+0.99639	0	0.003979	150
tau_b_rel_E (s)	+0.25	+0.24191	0	0.001242	150
W_eff (E src, signed)	+0.23267	+0.22838	0.07641	0.07504	15048
W_eff (I src, signed)	-0.31062	-0.30571	0.07724	0.07603	14977
W_eff (E src)	+0.23267	+0.22838	0.07641	0.07504	15048
W_eff (I src)	+0.31062	+0.30571	0.07724	0.07603	14977
W_in_eff (input neurons)	-0.00084237	-0.0009372	0.1008	0.09914	6675
c_E[j=0] (SFA coupling)	+0.05	+0.051808	0	0.0004028	150
c_E[j=1] (SFA coupling)	+0.05	+0.051854	0	0.0004949	150
c_E[j=2] (SFA coupling)	+0.05	+0.051913	0	0.0005438	150
c_0_E[j=0] (SFA offset)	+0	+0.034778	0	0.007287	150
c_0_E[j=1] (SFA offset)	+0	+0.034521	0	0.006991	150
c_0_E[j=2] (SFA offset)	+0	+0.033955	0	0.006951	150
a_0 (threshold)	+0.35	+0.37489	0	0.01633	300
readout_weight	+0.00078219	+0.00083932	0.06654	0.06087	6675
readout_bias	+0	+0.00013959	0	0.001394	89
ic.ic (per-variant init)	+0.10327	+0.10327	0.3829	0.3829	900

srnn-sfa-e-only-per-neuron

isp-rename-10e — effective time constants (srnn-sfa-e-only-per-neuron)

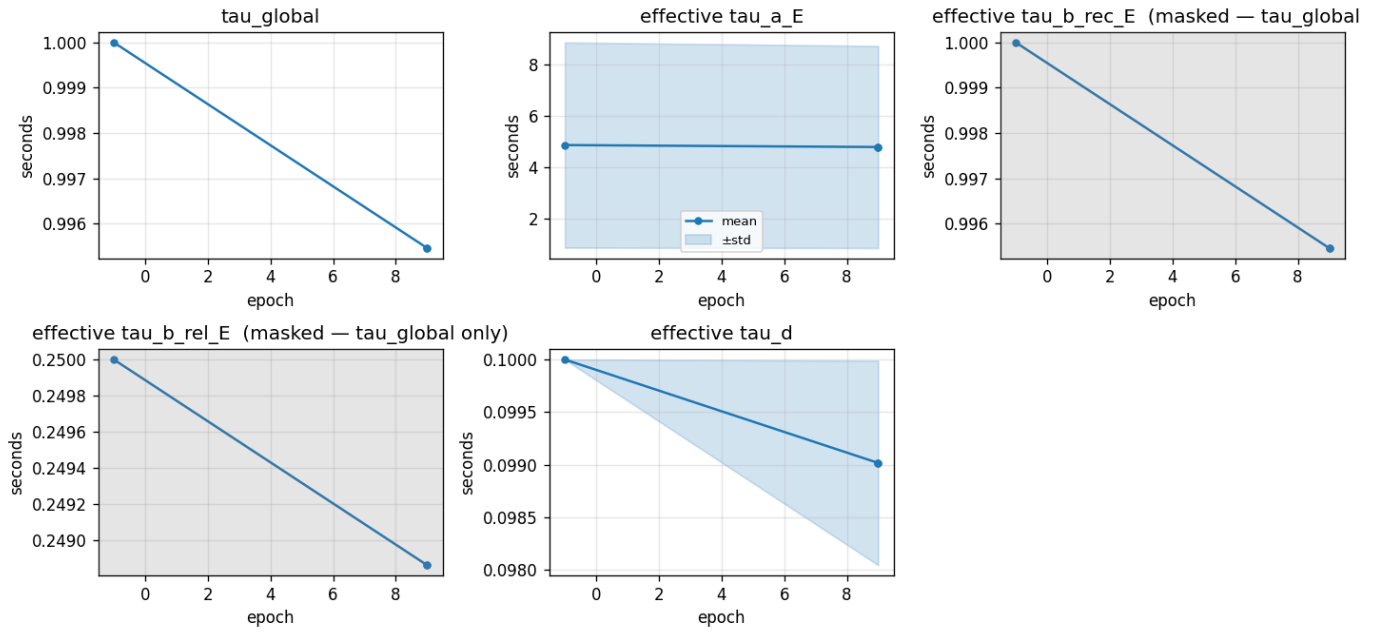


Figure 7: Tau evolution

isp-rename-10e — recurrent E vs I weights (srnn-sfa-e-only-per-neuron)

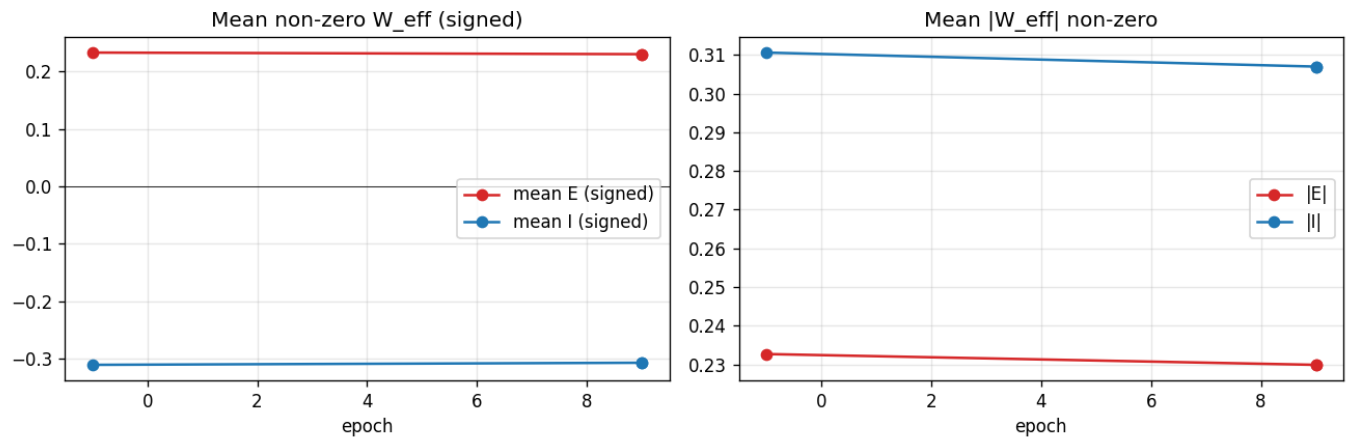


Figure 8: W_EI evolution

srnn-sfa-e-only-per-neuron — parameter table

```
# isp-rename-10e - variant srnn-sfa-e-only-per-neuron (k=2)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.99546	-	-	scalar
tau_d (s)	+0.1	+0.099016	0	0.0009727	300
tau_a_E[j=0] (s)	+0.25	+0.24492	0	0.001412	150
tau_a_E[j=1] (s)	+4.3832	+4.3127	0	0.004419	150
tau_a_E[j=2] (s)	+10	+9.8466	0	0.00305	150
tau_b_rec_E (s)	+1	+0.99546	0	0	150
tau_b_rel_E (s)	+0.25	+0.24886	0	0	150
W_eff (E src, signed)	+0.23267	+0.22988	0.07641	0.07553	15048
W_eff (I src, signed)	-0.31062	-0.307	0.07724	0.07634	14977
W_eff (E src)	+0.23267	+0.22988	0.07641	0.07553	15048
W_eff (I src)	+0.31062	+0.307	0.07724	0.07634	14977
W_in_eff (input neurons)	-0.00015382	-1.6796e-05	0.09934	0.0982	6675
c_E[j=0] (SFA coupling)	+0.05	+0.050858	0	0.0003478	150
c_E[j=1] (SFA coupling)	+0.05	+0.050909	0	0.0004038	150
c_E[j=2] (SFA coupling)	+0.05	+0.050954	0	0.0004537	150
c_0_E[j=0] (SFA offset)	+0	+0.018318	0	0.008675	150
c_0_E[j=1] (SFA offset)	+0	+0.018447	0	0.008812	150
c_0_E[j=2] (SFA offset)	+0	+0.018429	0	0.008721	150
a_0 (threshold)	+0.35	+0.3618	0	0.01082	300
readout_weight	-0.00098972	+0.00020923	0.06678	0.06622	6675
readout_bias	+0	+0.00052439	0	0.009408	89
ic.ic (per-variant init)	+0.13078	+0.13078	0.4172	0.4172	900

srnn-sfa-e-only-skip-per-neuron

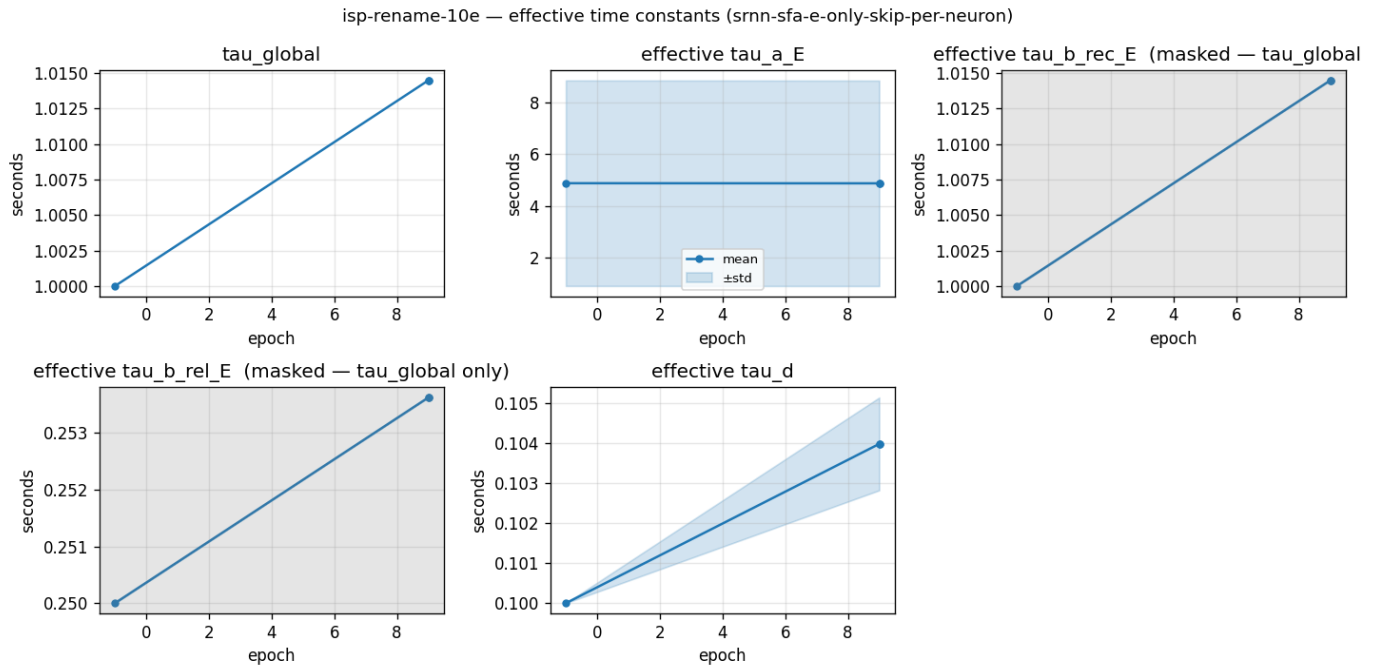
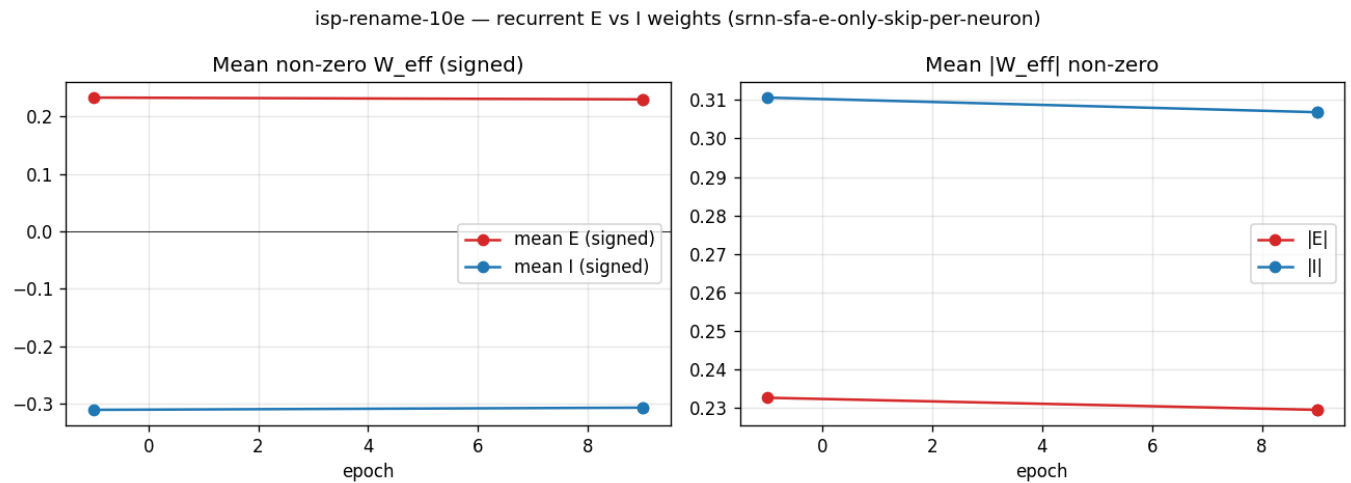


Figure 9: Tau evolution

Figure 10: W_{EI} evolution

srnn-sfa-e-only-skip-per-neuron — parameter table

```
# isp-rename-10e - variant srnn-sfa-e-only-skip-per-neuron (k=3)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+1.0145	-	-	scalar
tau_d (s)	+0.1	+0.10399	0	0.001161	300
tau_a_E[j=0] (s)	+0.25	+0.24775	0	0.001364	150
tau_a_E[j=1] (s)	+4.3832	+4.3767	0	0.004588	150
tau_a_E[j=2] (s)	+10	+9.9957	0	0.003787	150
tau_b_rec_E (s)	+1	+1.0145	0	0	150
tau_b_rel_E (s)	+0.25	+0.25362	0	0	150
W_eff (E src, signed)	+0.23267	+0.22954	0.07641	0.07541	15048
W_eff (I src, signed)	-0.31062	-0.30682	0.07724	0.0763	14977
W_eff (E src)	+0.23267	+0.22954	0.07641	0.07541	15048
W_eff (I src)	+0.31062	+0.30682	0.07724	0.0763	14977
W_in_eff (input neurons)	-0.0011621	-0.00088393	0.09962	0.09614	6675
c_E[j=0] (SFA coupling)	+0.05	+0.051154	0	0.0003285	150
c_E[j=1] (SFA coupling)	+0.05	+0.05129	0	0.0004489	150
c_E[j=2] (SFA coupling)	+0.05	+0.051399	0	0.0005162	150
c_0_E[j=0] (SFA offset)	+0	+0.028053	0	0.007471	150
c_0_E[j=1] (SFA offset)	+0	+0.028959	0	0.007489	150
c_0_E[j=2] (SFA offset)	+0	+0.028963	0	0.007423	150
a_0 (threshold)	+0.35	+0.36842	0	0.01321	300
readout_weight	+0.00016801	+0.00019328	0.0673	0.06327	6675
readout_bias	+0	+3.8623e-05	0	0.006911	89
ic.ic (per-variant init)	+0.09231	+0.09231	0.5294	0.5294	900

srnn-std-e-only-per-neuron

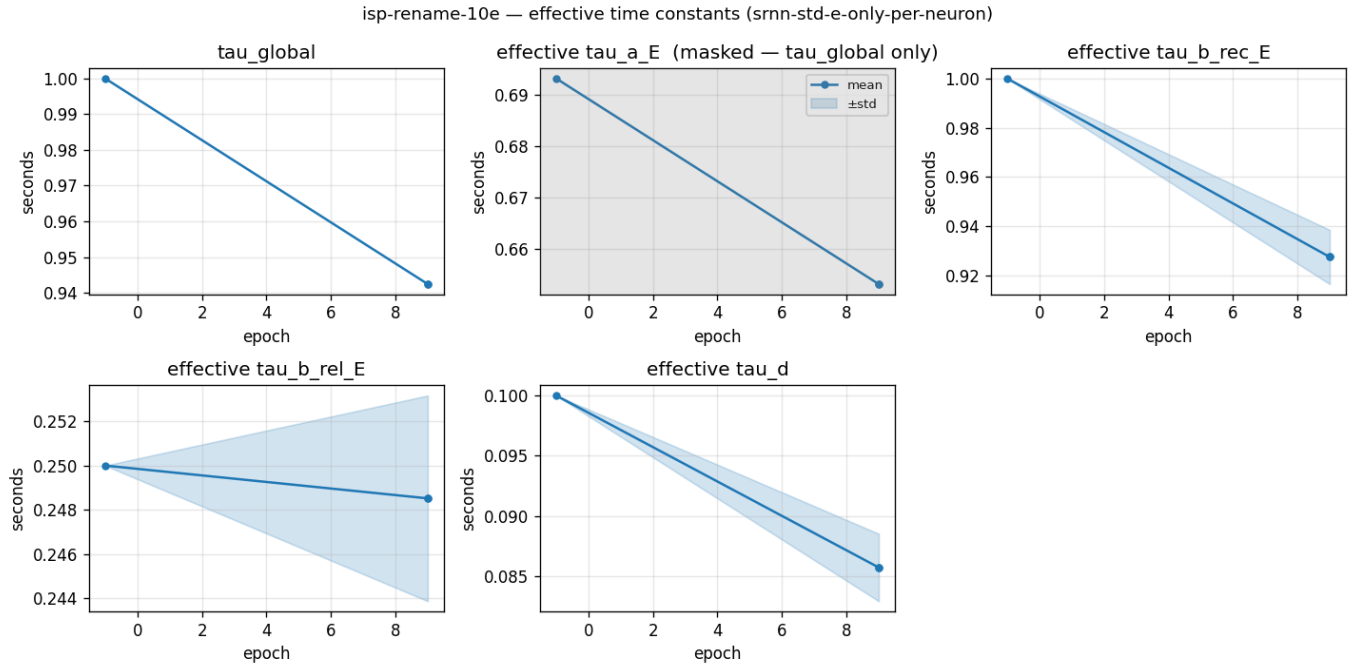


Figure 11: Tau evolution

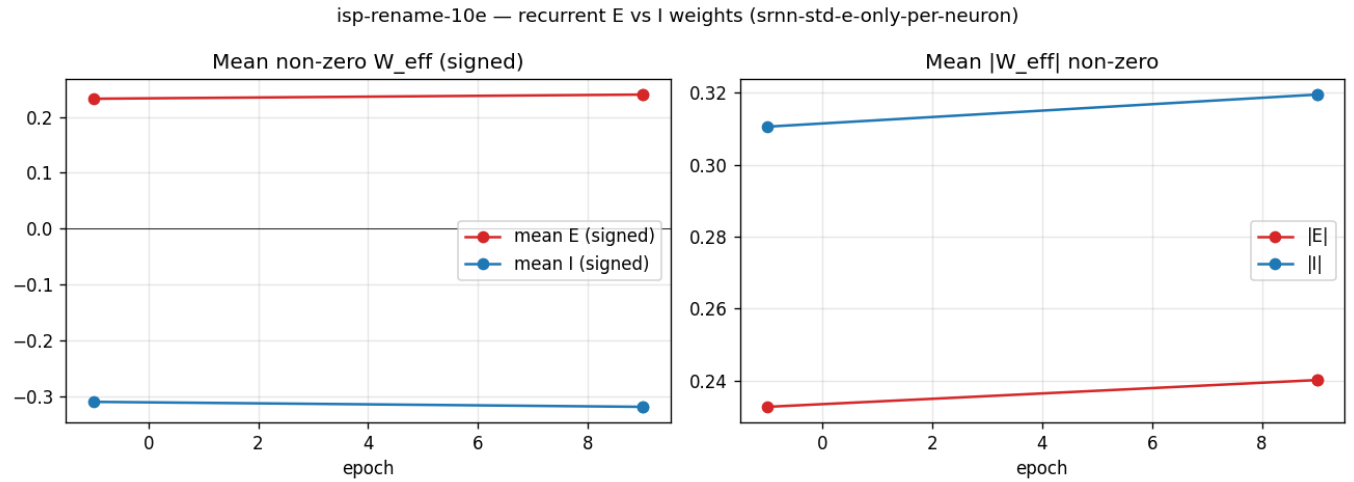


Figure 12: W_EI evolution

srnn-std-e-only-per-neuron — parameter table

```
# isp-rename-10e - variant srnn-std-e-only-per-neuron (k=4)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.94245	-	-	scalar
tau_d (s)	+0.1	+0.085709	0	0.002801	300
tau_b_rec_E (s)	+1	+0.92759	0	0.01098	150
tau_b_rel_E (s)	+0.25	+0.24852	0	0.004655	150
W_eff (E src, signed)	+0.23267	+0.24012	0.07641	0.07896	15048
W_eff (I src, signed)	-0.31062	-0.31955	0.07724	0.07956	14977
W_eff (E src)	+0.23267	+0.24012	0.07641	0.07896	15048
W_eff (I src)	+0.31062	+0.31955	0.07724	0.07956	14977
W_in_eff (input neurons)	+0.00019944	+0.00036307	0.09994	0.1077	6675
a_0 (threshold)	+0.35	+0.32272	0	0.02254	300
readout_weight	+0.00074661	+0.0024107	0.06657	0.07225	6675
readout_bias	+0	-0.00404	0	0.02586	89
ic.ic (per-variant init)	+0.090819	+0.090819	0.3835	0.3835	900

srnn-std-e-only-skip-per-neuron

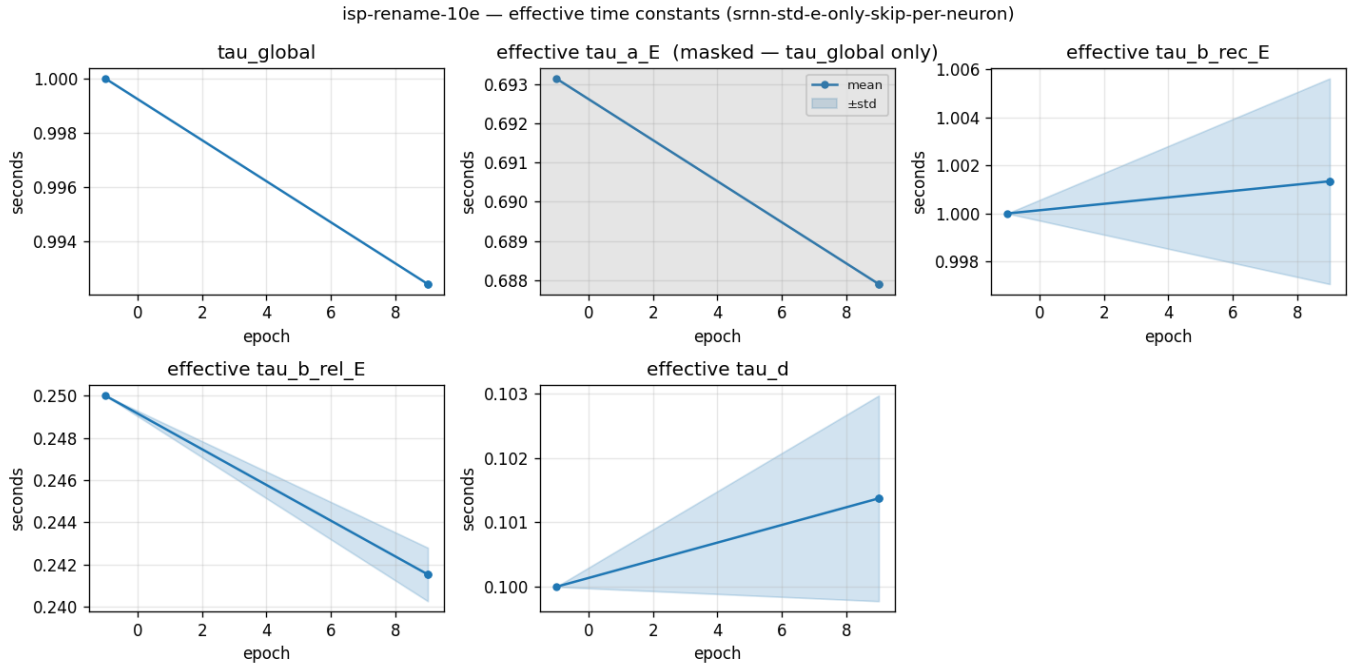


Figure 13: Tau evolution

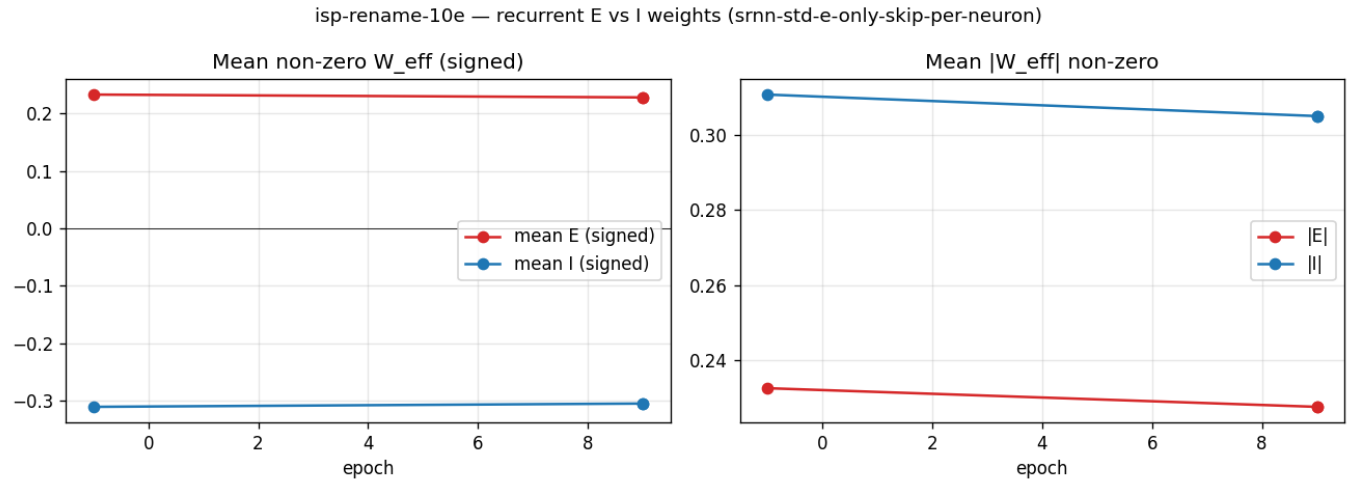


Figure 14: W_EI evolution

srnn-std-e-only-skip-per-neuron — parameter table

```
# isp-rename-10e - variant srnn-std-e-only-skip-per-neuron (k=5)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.99243	-	-	scalar
tau_d (s)	+0.1	+0.10137	0	0.001597	300
tau_b_rec_E (s)	+1	+1.0013	0	0.00428	150
tau_b_rel_E (s)	+0.25	+0.24153	0	0.001266	150
W_eff (E src, signed)	+0.23267	+0.22772	0.07641	0.07482	15048
W_eff (I src, signed)	-0.31062	-0.30492	0.07724	0.07584	14977
W_eff (E src)	+0.23267	+0.22772	0.07641	0.07482	15048
W_eff (I src)	+0.31062	+0.30492	0.07724	0.07584	14977
W_in_eff (input neurons)	+0.00036119	+0.00034556	0.09956	0.09879	6675
a_0 (threshold)	+0.35	+0.37506	0	0.01671	300
readout_weight	+0.00055372	-0.00031886	0.06572	0.05966	6675
readout_bias	+0	-0.00033867	0	0.002078	89
ic.ic (per-variant init)	+0.083543	+0.083543	0.384	0.384	900